

EDUCATION

Texas A&M University Ph.D. in Statistics	College Station, TX 2020–Present
University of Colorado Boulder M.S. in Applied Mathematics	Boulder, CO 2017–2018
University of Colorado Boulder B.S. in Applied Mathematics	Boulder, CO 2014–2018

EXPERIENCE

Texas A&M University Research Assistant	College Station, TX 2020–Present
– Vecchia Approximation for Bayesian Optimization – Using Gaussian processes approximation for Bayesian optimization.	
National Institute of Standards and Technology (NIST) Mathematical Statistician	Boulder, CO 2019–2020
– Hierarchical Gaussian Processes for Outlier Detection – Created an outlier detection procedure using Gaussian processes. – Permutation Test for Forensic Analysis of Glass – Developed a permutation test for μ-XRF spectra to determine if two sets of glass come from the same source. – Generative Adversarial Networks on Low Dimensional Data – Investigated the impact of data quality and complexity on fidelity of samples generated by GANs.	
National Institute of Standards and Technology (NIST) Mathematical Statistician	Boulder, CO 2019–2020
– Hierarchical Gaussian Processes for Outlier Detection – Created an outlier detection procedure using Gaussian processes. – Permutation Test for Forensic Analysis of Glass – Developed a permutation test for μ-XRF spectra to determine if two sets of glass come from the same source. – Generative Adversarial Networks on Low Dimensional Data – Investigated the impact of data quality and complexity on fidelity of samples generated by GANs.	
University of Colorado Boulder/NIST Professional Research Assistant	Boulder, CO 2018–2019
– See Above.	
University of Colorado Boulder Research Assistant	Boulder, CO 2015–2018
– Statistics Consulting Lab – Acted as consultant for groups in Environmental Engineering, Mechanical Engineering and Academic Services. – Applied Math Department – Undergraduate research assistant for project between physics and applied math.	

PUBLICATIONS

- [1] **F. Jimenez**, A. Koepke, and M. Frey, “Generative Adversarial Network Performance for Low Dimensional Data”, Submitted to NIST Journal of Research.
- [2] A. Koepke, **F. Jimenez**, K. Kroenlein, and C. Muzny, “Hierarchical Bayesian change point models for chemical properties inference”, presented at the 2019 Joint Statistical Meetings, 2019.
- [3] J. Splett, A. Koepke, and **F. Jimenez**, “Estimating the Parameters of Circles and Ellipses Using Orthogonal Distance Regression and Bayesian Errors-in-Variables Regression”, presented at the 2019 Joint Statistical Meetings, 2019.
- [4] K. Tucker, B. Zhu, R. J. Lewis-Swan, J. Marino, **F. Jimenez**, J. G. Restrepo, and A. M. Rey, “Shattered time: Can a dissipative time crystal survive many-body correlations?”, *New J.Phys.*, vol. 20, Dec. 2018.

TEACHING

- **Teaching Assistant** at Texas A&M University Fall 2020
Statistical Learning (Stat 436)
- **Teaching Assistant** at University of Colorado Boulder Fall 2017
Calculus 1 (APPM 1350)
- **Short Course Instructor** at University of Colorado Boulder Fall 2017 - Spring 2018
Intro. to R and Social Network Analysis
- **Learning Assistant** at University of Colorado Boulder 2016-2017
Matrix Methods and Physics I
- **Instructing Assistant with SASC** at University of Colorado Boulder 2015-2016
College Algebra and Intro. to College Math

PROGRAMMING LANGUAGES

- **Advanced:** Python, R.
- **Proficient:** C++.
- **Beginner:** Java, SQL, Scala.

TOOLS & LIBRARIES

- **Advanced:** Numpy, ggplot, Pytorch.
- **Intermediate:** Linux, Spark, Slurm, scikit-learn.
- **Beginner:** Bash, SQL, AWS.

TALKS

- Hierarchical GPs for Outlier Detection 2020
JSM, Topic Contributed
- Glass Stats and Crime: Let’s Make the Right Decision 2019
JSM, Topic Contributed
- Generative Adversarial Network Performance for Low Dimensional Data 2019
QRPC, Topic Contributed
- Generative Adversarial Network Performance for Low Dimensional Data 2019
NIST Colloquium

POSTERS

- Measuring Inter-operator Variability 2016
Rocky Mountain Fluid Mechanics Symposium
- Interacting Quantum Dipoles 2016
Dynamics Days

SCHOLARSHIPS AND AWARDS

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|---|------|
| • Lechner Graduate Grant, \$6,000 | 2020 |
| • Cash-In-Your-Account Award, \$200 | 2020 |
| • Biological Sciences Initiative, \$700 | 2016 |
| • Biological Sciences Initiative, \$3,500 | 2015 |